

PLATFORM COMPETITION IN TWO-SIDED MARKETS: EVIDENCE FROM THE PC
VIDEO GAME MARKET

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ABSTRACT

Do entrenched platforms in two-sided markets face price competition when a well-funded challenger enters the market and subsidizes both sides? This paper examines the launch of the Epic Games Store and observes the impact their entry had on the dominant platform, Steam, in the PC game market. Using daily price and sale data of 900 Steam games from 2013-2024, I analyze the impact of Epic's entry at two main levels. First, the impact is observed at the market level with an event-study regression comparing games that list on both platforms against games that list solely on Steam, and I find that prices declined while discounting grew more frequent and deeper over the sample period, though this design is descriptive and cannot attribute those changes to Epic's entry. Second, the impact of Epic's platform listing a game on the pricing and discounting behavior of the same title on Steam is observed in the game-level portion of this study with two separate difference-in-differences models, and I find that Epic's listings caused prices to rise on Steam by ~\$0.50 within 90 days and discounting depth to increase ~1.2 percentage points within 180 days. All other results of the game-analysis section are null and collectively support the claim that Epic had little to no effect on Steam's pricing and discounting behavior when listing the same title. These results are consistent with two-sided market theory that claims steering transactions to an entrant platform is weak when users usage-singlehome on a dominant platform.

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CHAPTER 1: INTRODUCTION

In December 2018, Epic Games launched a storefront to challenge Steam's dominance over PC video game distribution with an aggressive strategy including \$2B in promotional spending for buyers and significantly better revenue cuts for game developers. By 2025, the Epic Games Store grew to the second largest game launcher on PC with 67M monthly active users (Dubois, 2026). Epic successfully created a competing platform in a market where Steam dominated for 10+ years without any notable challenger. A well-funded challenger of this size, subsidizing both buyers and sellers, might be expected to pressure the dominant incumbent (Steam) to compete on prices, but the economics of network-driven platforms suggest that incumbent firms should retain their dominance and remain insulated from this type of competitive pressure. The impact of this entry on the pricing and discounting behavior of sellers on the incumbent platform is the central question of this study.

The structure of these platforms in the PC game market makes this a strong empirical setting to test this research question. Game publishers utilize platforms like Steam and the Epic Games Store to distribute their video games while maintaining full autonomy over their prices. In addition to this, publishers can freely enter and exit these platforms or even multihome. Because publishers set prices, Steam's competitive response appears directly in the prices and discounting of Steam games and is observable at two main levels. First is at the market-level which examines aggregate price and sale trends across the market, and the second is at the game-level which observes price and

sale adjustments when a title already on Steam is listed on Epic. The analysis for this study is separated into these two sections.

In order to observe Steam's reaction to Epic's entry, daily pricing and sales data is collected for 900 Steam games from 2013 to 2024. With this data, the two main levels of analysis are performed. First, prices and sales on Steam are observed in the total data sample period (2013 – 2024) to observe the market-level results around Epic's entry into the market. The primary model from this section includes an event-study regression comparing the trends of games only sold on Steam and games that are either currently sold on both platforms or will be in the future. This section gives insight into how Epic's entry impacted prices and sales on Steam in the long run. The second level of analysis is at the game-level. This section observes the impact of a game listing on the Epic Games Store on that same title's Steam price and sale composition within smaller time windows to test the short-run effect. The primary models in this section include stacked and synthetic control difference-in-differences models. Due to the inclusion of control games that match game characteristics and pre-treatment trend levels, the difference-in-differences models provide the most causal results of the whole study. Throughout the entire study, price, inflation-adjusted price, sale frequency, and discount depth are the outcome variables tested.

Descriptive statistics on Steam-only and future / current dual-platform games show a pre-existing, statistically significant difference in all outcome variables, except discount depth. Additionally, the distribution of release dates and genres between the two groups display some significant differences, implying selection into listing rather than random assignment. As a result of this and the long time window, all market-level results

must be interpreted as descriptive and not a causal statement on Epic's entry into the market. Regardless, this section indicates that prices have decreased and sale frequency and depth have increased for both groups in the long run. These results are slightly stronger for games classified as dual platform and are only significant a few years after Epic's entry (2022-2024). On the game-level, the difference-in-differences models show a mostly insignificant effect of Epic's listings on prices and sales of the same game on Steam. The two significant results are a ~\$0.50 increase in the price on Steam within 90 days of the same game launching on Epic, which becomes insignificant 180 days post-launch, and the discount depth increasing 1.2 percentage points within 180 days. Therefore, the market-level results indicate a small downward drift in prices and upward drift in sale frequency and depth in the long run. Meanwhile, the game-level results find no downward pressure on prices within the short run of a game listing on Epic, but a price and discount depth increase within 90 and 180 days of the listing, respectively.

An extensive background on relevant two-sided market and network externality literature is conducted in this study to theorize expected results and compare the findings to these literature-backed expectations. These referenced papers are organized into foundational, entry and dominance, and same-market categories and provide the theoretical framework and empirical context for studying the impact of Epic's entry into the market on Steam's pricing and discounting behavior. Relevant literature points to two-sided markets with existing network externalities and access to multihoming creating an asymmetric equilibrium with a dominant platform that benefits both consumers and producers. In addition to this, the literature on entry in these markets alludes to entrants having a difficult time capturing market share. As a result, this literature defines the PC

game market as a two-sided market with network externalities that are expected to insulate the dominant incumbent (Steam) from competitive entry, and the expected impact of Epic's entry is null in the short run and minimal in the long run. The most directly related empirical work, Jeong et al. (2026), examines the spillover effects of Epic's free-game promotional events on Steam revenue in the same market, but the effect of a game's listing on pricing and discounting behavior on Steam remains untested. This study addresses that gap.

The remainder of this paper is organized as follows. Chapter 2 provides background in two main sections: an industry background on the PC game market and its unique features, and a review of relevant two-sided market and network externality literature. Chapter 3 builds on the industry background and literature review to construct and explain the central hypotheses of this study, and defines the market within the scope of related literature. Chapter 4 gives a detailed background on the data collected and used in this study. Chapter 5 describes the methodology behind the market- and game-level analysis. Chapter 6 presents the descriptive statistics, market-level and game-level results of the study. Chapter 7 summarizes the results, examines the outcome of each previously specified hypothesis, and details areas of future research and the limitations of this study. Lastly, Chapter 8 concludes.

CHAPTER 2: BACKGROUND

2.1 Industry Background

PC gaming accounts for ~20% of the \$190B global video game industry with ~90% of PC game distribution occurring on game launchers like Steam and Epic (Jeong et al., 2026). In contrast to gaming consoles, such as Xbox or Nintendo, PC gaming does not have a hardware gatekeeper determining which games can be sold on that system. Rather, anyone is capable of creating, selling, and distributing their own game on PC. However, despite game publishers having the freedom to distribute their game independently, most opt into using game launchers to reach a bigger audience, since platforms like Steam and Epic already have a large group of buyers willing to purchase games. Additionally, game publishers rely on these game launchers to expedite certain processes, such as game updates and multiplayer features, while also dealing with piracy prevention through DRM. As a result, game launchers host a large population of sellers who utilize the platform as their online storefront to reach buyers.

Given that these game launchers operate as central online storefronts for PC gaming, the success of a platform relies on the number of participants on the buyer and seller-side. Publishers are more likely to join platforms with a larger pre-existing group of buyers, and buyers are more likely to shop on and utilize platforms that offer a wide variety of games. In other words, game launchers operate in a market where network externalities exist and dictate who becomes successful. In addition to this, game publishers have full autonomy over the price and discounts of their games. Game

launchers have no control over how a game developer prices their game or organizes discounts, but platforms like Steam and Epic do try to encourage participation into seasonal sales by promoting it as a marketing opportunity to game developers. Given the lack of control over prices, game launchers instead generate revenue through revenue sharing models where they take a percentage of each sale a publisher makes on the platform. Platform revenue is generated solely from the seller-side, with buyers having free access to these game launchers and paying no entry or membership fee.

Steam launched on September 12, 2003 and allowed third-party games to be sold on the platform in 2005. Since then, Steam has operated as the dominant and primary game launcher in the PC gaming market. Outside of Steam, game publishers were required to sell their games directly to consumers or create their own game launchers. Publisher-specific launchers did exist and compete with Steam directly, but never or rarely sold third-party games and only existed for major game developer companies with the ability to operate and maintain a game launcher. As a result, Steam spent 10-15 years mostly uncontested as the dominant game launcher in this market.

On December 6, 2018, Epic Games launched its own game platform called the Epic Games Store with the primary intention of competing with Steam and stealing market share. It was the first major platform outside of Steam to extend its store to third-party games and allow any game publishers to list their games on the platform. With the intention of capturing Steam's buyers and sellers, Epic utilized multiple competitive tools on both the buyer and seller-side. For buyers and gamers, Epic offered over \$2B in promotional spending, primarily in the form of weekly free games to registered users, platform-exclusive titles, and its Epic Rewards cashback program (Dubois, 2026).

However, Epic utilized its strongest competitive tool on the seller-side by offering a significantly lower revenue sharing cut. Epic's revenue cut is only 12%, giving publishers 88% of their game's all-time revenue. This was significantly lower than Steam's 30% revenue cut which it used for most of its existence. Potentially in response to Epic's entry in the market, Steam restructured its revenue sharing model days before the Epic Games Store launched with a tiered system that lowered the revenue cut to 25% when all-time revenue passed \$10M and 20% at \$50M. Epic also made a recent adjustment to its revenue cut in June 2025 making the first \$1M in annual revenue go 100% to the developers, but this change occurred after the sample period of this study.

As a result of its aggressive entry, Epic has become the second-largest game launcher in the PC game market with 317M registered accounts as of 2025 (Dubois, 2026). Despite Epic's success, Steam still operates as the dominant platform based on all metrics. The clearest sign is the gap between users and spending. The Epic Games Store draws 35%-40% of PC monthly active users but only accounts for 5%-8% of market spending (Packwood, 2026). In 2025, Steam averaged 132M monthly active users on its platforms, and Epic averaged 67M (Dubois, 2026). In that same year, Steam generated ~\$16.9B in revenue compared to the Epic Games Store's \$1.16B (Dubois, 2026).

Another area where the Epic Games Store appears to struggle is the total number of games in each platform's catalog. As of 2025, Steam contains 117,000+ games in its catalog while the Epic Games Store has only 6000+ (Dubois, 2026). Steam's advantage on both the buyer and seller side through more spending and a larger catalog confirms the dominant incumbent position against which Epic's entry is measured.

2.2 Literature Review

This review provides the theoretical background and empirical context for studying whether Epic's entry affected Steam's pricing and discounting. The first four papers of this review establish the foundations of two-sided market and network externality literature. Katz and Shapiro (1985) introduce network externalities and the expectations that drive dominance, while Rochet and Tirole (2003, 2006) and Armstrong (2006) develop the pricing framework for two-sided markets, including the distinction between singlehoming and multihoming participants. Caillaud and Jullien (2003) examine the entry dynamics of two-sided markets. This is followed by Ko and Shen's (2021) paper on dominance in two-sided markets and its welfare implications. Lastly, this section concludes with a recent, same-market paper by Jeong et al. (2026), which studies the effect of Epic's weekly free-game promotional event on Steam sales.

Katz and Shapiro (1985) provide one of the earliest and most foundational papers on network externalities. The authors characterize network externalities as arising when "the utility that a user derives from consumption of the good increases with the number of other agents consuming the good" (Katz & Shapiro, 1985, p. 424). The key finding of this paper, as it relates to this study, is that when network externalities exist and consumers expect a firm to be dominant, that firm then becomes dominant. This self-fulfilling dominance arises from consumers' increased willingness to spend on products they expect will attract a large network of other users. Within the scope of this study, this finding highlights the mechanism through which platforms, like Steam, dominate certain markets. If consumers perceive Steam as the market leader, their willingness to purchase games on the platform may be higher than for competing platforms.

Rochet and Tirole (2003) provide one of the founding papers on two-sided markets. They find that most markets with network externalities fall under the definition of two-sided markets and that platforms in these markets must choose a price structure rather than simply a price level. The distinction between the two is that the price level refers to the total extracted from buyers and sellers, while the price structure refers to how the total is divided between those two groups. Rochet and Tirole also define the singlehoming index, which measures the loyalty of a platform's users as the portion of users who would stop transacting if that platform were unavailable. The index equals one when all users singlehome and falls as multihoming becomes more prevalent. When one side of the market multihomes, that side can route its transactions to whichever platform is cheaper, a strategy the authors term "steering." Because steering allows a platform to capture transactions by undercutting its rival, price competition intensifies as multihoming rises and the singlehoming index falls. Applied to this study, steering requires that the multihoming side capable of routing transactions actually does so. If buyers transact primarily or entirely on Steam, the singlehoming index remains high and steering is weak, making Epic's lower fees exert little downward pressure on Steam's prices.

Armstrong (2006) offers the other foundational background on two-sided market competition. He identifies three main determinants of equilibrium prices, consisting of the relative size of cross-group externalities, whether platforms charge per-transaction or lump-sum fees, and whether participants singlehome or multihome. Around these factors, he develops three models, covering a monopoly platform, competing platforms where both sides singlehome, and a competitive bottleneck where one side multihomes and the

other singlehomes. In the competitive bottleneck model, each platform holds a "local monopoly" over access to its singlehoming users, because the multihoming side must go through that platform to reach them. Because game publishers multihome, the competitive bottleneck model is the one that fits this market best. The two-sided singlehoming model serves as a counterfactual where entry would be expected to lower prices. Armstrong finds that this singlehoming model leads to increased price competition, meaning if Epic's entry exerted no downward pressure on Steam's prices, then the PC game market is better described by the competitive bottleneck model. Under the bottleneck model, Steam's local monopoly over its singlehoming buyers means Epic's lower fees cannot reach the transactions that continue to occur on Steam.

The last foundational paper included in this literature review is Rochet and Tirole's (2006) progress report that unifies their usage-based framework with Armstrong's membership-based approach. First, the paper gives a stronger definition of two-sided markets as markets where the price structure affects the transaction volume independently of the price level. This distinction between price level and price structure is expanded to find that the competitive decisions made on one side of the market are distinct from the decisions on the other side, despite being set jointly. However, the main finding as it relates to this study is how homing status can differ between usage and membership. They find that users can multihome their membership across multiple platforms while singlehoming their usage on one platform. Because steering works through the routing of transactions, usage-singlehoming leaves a rival platform with few transactions to attract, weakening the price competition that steering would otherwise create. This idea complements Armstrong's theory of platforms having a local monopoly

over their singlehoming users. These two findings strongly relate to the PC game market. The price structure versus price level findings indicate that competition on the seller-side through revenue sharing cuts does not automatically imply that effects will appear on the buyer side. Additionally, the membership versus usage distinction may apply to Steam and Epic given the gap between Epic's user share and its share of spending. Despite having a 35%-40% share of monthly active users, Epic still only accounts for 5%-8% of spending in the market, implying that buyers may multihome membership but singlehome their usage on Steam (Packwood, 2026).

Caillaud and Jullien (2003) shift the focus from pricing within established two-sided markets to the dynamics of entry into them. One of the key findings of this paper as it relates to this study is the “divide-and-conquer” strategy for platforms to capture market share. This strategy involves subsidizing the participation of one side of the market, “divide,” and recovering losses through the other side, “conquer.” The authors find that when multihoming exists, it is easy for platforms to “divide” through subsidization, but difficult to pull the entire transaction over to the platform. They expand on this by defining dominant incumbent firms as the “first source” that most transactions operate through and smaller entrants as the “second source” that only capture transactions the dominant platform missed. This framing fits Epic’s entry and offers a mechanism consistent with the observed gap between Steam’s and Epic’s revenue. Essentially, Epic can easily “divide” and pull customers to its platform through subsidization because of the access to multihoming, but it is hard to “conquer” and capture the transaction from the “first source” (Steam).

Ko and Shen (2021) extend this literature by characterizing two-sided markets through the strength of network effects. They show that market structure is defined by a network-effect parameter β , which creates three types of equilibria. When network effects are weak ($\beta \leq 0.5$), a symmetric equilibrium exists in which platforms share the market equally. Moderate network effects ($0.5 < \beta < 1$) create an asymmetric equilibrium with one dominant platform alongside a smaller competitor. Strong network effects ($\beta \geq 1$) create a monopoly equilibrium in which a single platform serves the entire market. They further show that once β exceeds 0.5, the symmetric equilibrium becomes unstable, since a group of sellers could jointly benefit by moving to the same platform. Only the asymmetric and monopoly equilibria are protected from this kind of coordinated deviation. Therefore, dominance is a stable outcome rather than just a possible one. Their central finding is that an asymmetric, dominant-platform equilibrium always yields higher consumer surplus, producer surplus, and total welfare than a symmetric equilibrium where platforms split the market evenly. In a monopoly equilibrium, this effect continues for producer surplus and total welfare, but consumer surplus only rises when $1 \leq \beta < 1.1$ and falls otherwise. Within their model, a platform's dominance does not harm consumers, but a pure monopoly can.

Ko and Shen's baseline model assumes that platforms charge buyers a fixed fee and that both sides of the market singlehome. However, their online appendix shows these results are robust to platforms charging sellers a revenue share and to sellers multihoming while buyers singlehome. The online appendix also allows sellers to differ in the cost of listing on both platforms and finds that in equilibrium sellers who multihome are those with the lowest multihoming costs. Therefore, the group of sellers

that multihomes is selected based on costs rather than by random selection. Together, these findings provide insight into Steam's dominance in this market and Epic's limited success in capturing transactions.

Lastly, Jeong et al. (2026) provide the closest same-market evidence to this study. They examine the effect of Epic's weekly free-game promotion on Steam sales of the same title. Their primary finding is that Steam sales rose by 59.2% during the promotional week, despite the title being offered for free on Epic. In a survey performed to uncover the reasoning behind this counterintuitive occurrence, the authors found that buyers opt to purchase the game on Steam due to network externalities, constrained by the gap in price between the platforms. Multiplayer features and access to additional content were the largest drivers behind users purchasing the game on Steam. The authors also report that Epic selects high-potential titles for these promotions rather than selecting at random from its catalog.

In addition to their short-run finding, Jeong et al. (2026) report a long-run finding on the impact of Epic's promotional activities on Steam. Over a two-year window around these promotions, Steam consumers became more price-elastic and discounting on these games grew more frequent and deeper. Because this result is driven by Epic's repeated promotional events rather than a title listing on the platform and is measured over a longer time horizon, these findings answer a different question from the short-run listing effect this study estimates. However, it does motivate the expectation that sustained competitive pressure may affect Steam's pricing over a longer period. Although Jeong et al. (2026) research a separate question from this study, their findings provide insight into the dynamics of the PC game market. Their results support Steam's "first source" status

and local monopoly over usage-singlehoming buyers, and they indicate the limitations of Epic’s “divide” strategy, given that buyers continue to purchase on Steam when the title is free on Epic.

Collectively, these papers create the theoretical background and empirical context for this study. Katz and Shapiro (1985) explain why an incumbent platform remains entrenched, as consumer expectations about which platform will dominate are self-fulfilling. Rochet and Tirole (2003, 2006) and Armstrong (2006) explain how that entrenchment shows up in prices. Because steering requires the multihoming side to route transactions to the cheaper platform, an incumbent whose buyers multihome in membership but singlehome in usage retains a local monopoly over the bulk of transactions. Caillaud and Jullien (2003) explain why an aggressive entry strategy may not translate into downward pressure on prices, since multihoming makes an entrant easy to “divide” with but difficult to “conquer” through. Ko and Shen (2021) explain why the resulting dominance is a stable equilibrium and why it can benefit consumers. Jeong et al. (2026) demonstrate in their same-market study that Epic’s most aggressive buyer-side competitive tool failed to draw sales away from Steam.

One caveat applies across the theoretical work. Each of these models describes a static equilibrium rather than the path of adjustment following entry, so none predicts a specific short-run price response. Their role here is to establish why an entrenched, network-effect-rich incumbent with usage-singlehoming buyers would be expected to remain insulated from a challenger's entry. The following chapter builds on this expectation to develop the study's three hypotheses.

CHAPTER 3: CONCEPTUAL FRAMEWORK AND HYPOTHESES

3.1 Market Characterization

Before creating the literature-backed hypotheses that the study tests, I first lay out the characteristics of the PC game market within the scope of the literature. First, the PC game market resembles a two-sided market where game launchers, such as Steam and the Epic Games Store, serve as intermediaries between buyers and game publishers. Both buyers' and sellers' value from these platforms relies on the participation of the other side, which is the defining feature of a two-sided market (Rochet & Tirole, 2003). Second, the observed structure of this market, in which a single dominant platform coexists with a smaller competitor, is consistent with moderate network effects, placing it in the β range where a dominant-platform equilibrium is stable and mutually beneficial for both consumers and producers (Ko & Shen, 2021). This then creates an asymmetric equilibrium in which Steam operates as the "first source" that transactions route through by default, while Epic operates as the "second source" capturing the transactions Steam misses (Caillaud & Jullien, 2003). As for the two sides of this market, sellers multihome across the two platforms, or can easily do so, while buyers membership-multihome but usage-singlehome on Steam (Rochet & Tirole, 2006). Therefore, Steam acts as the dominant platform in the market and holds a local monopoly over its usage-singlehoming buyers, which resembles the competitive bottleneck model developed by Armstrong (2006).

The study's three main hypotheses rely on this characterization of the market. Hypothesis I is motivated by the literature on usage-singlehoming and platforms maintaining a local monopoly over singlehoming users. Hypothesis II is motivated by the selection of sellers into multihoming described in Ko and Shen's (2021) online appendix and the non-random selection of titles documented by Jeong et al. (2026). Lastly, hypothesis III is motivated by the potential erosion of singlehoming local monopolies.

3.2 Hypothesis I: Short-Run Incumbent Insulation

The first hypothesis claims that a game listing on Epic's platform has no effect on that same title's price, sale frequency, or discount depth on Steam in the short run. Therefore, when a game lists on the Epic Games Store, price and sale composition should remain unchanged through a small time window. Since publishers, not the platforms, set the price of their games, a listing cannot pressure Steam's price through direct competition between the two platforms. Therefore, the listing can only affect the price on Steam by impacting the publisher's own Steam-side pricing problem. First, Epic's lower revenue cut (12% versus 30%) changes the publisher's margin, which could shift their pricing or discounting behavior on Steam. Second, a listing on Epic could shift demand on Steam through the additional exposure the title receives, leading to re-optimization of pricing and discounting on Steam. Steam's revenue share and platform terms are set market-wide and do not respond to an individual title's listing, meaning no channel runs through Steam's terms. However, the literature provides several reasons to expect both of these channels to be second-order in the short run.

Steam's market leader status partially exists due to expected dominance creating actual dominance in markets with network externalities (Katz & Shapiro, 1985).

Publishers expect the audience and buyers expect the catalog of games on Steam, leading to this self-fulfilling dominance. An entrant does not disrupt the equilibrium simply by existing. Rather, price discipline would have to operate through steering, requiring the multihoming side (publishers) to route transactions to the cheaper platform. Although publishers list on both stores, buyers determine where a purchase actually occurs. When buyers usage-singlehome on Steam, the transaction stays on Steam and steering is weak (Rochet & Tirole, 2003, 2006). With the Steam demand curve largely unaffected by Epic's availability, the publisher's optimal Steam price remains unchanged. As for Epic's revenue cut advantage, this competition occurs on the seller side, but the outcomes tested in this study reside on the buyer side. These are distinct sides of the price structure that are set jointly but remain separate levers (Rochet & Tirole, 2006). Given Steam's local monopoly over usage-singlehoming buyers, Steam's buyer-side prices are not competed down by Epic's presence (Armstrong, 2006). Epic can implement the "divide-and-conquer" strategy, which it likely intended with its \$2 billion in promotional spending, but "conquering" by capturing the transaction is difficult under multihoming (Caillaud & Jullien, 2003). With the expected network effects being moderate, dominance by a single platform is the anticipated stable outcome, meaning a single entry into the market should not be expected to alter that (Ko & Shen, 2021). As for the related same-market literature, Jeong et al. (2026) found that Epic's aggressive promotional strategy of offering free games to users not only failed to hurt Steam sales of the same game, but actually boosted sales by 59.2% within the same week.

This hypothesis is primarily tested in the game-level analysis section of this study. All three models in this section provide estimated differences in the outcome variables

between the pre- and post-listing periods of each individual game in the sample that lists on the Epic Games Store. Null results across these models would be consistent with this hypothesis.

3.3 Hypothesis II: Selection-Induced Differences

The second hypothesis states that observed differences in prices and sales between games offered solely on Steam and games offered on both platforms reflect selection into listing rather than a causal effect of listing. Essentially, the games that list on Epic are not randomly selected and may differ from Steam-only games for reasons that predate any listing. Unlike the first hypothesis's goal of observing a treatment effect, this hypothesis examines how differences between the two groups should be interpreted.

In a competitive bottleneck setting where sellers can multihome across both platforms, sellers differ in their cost of doing so (Ko & Shen, 2021, online appendix). In equilibrium, the sellers who multihome are those with low multihoming costs, meaning the dual-platform group is a systematically selected subset rather than a random sample (Ko & Shen, 2021, online appendix). In addition to this, Jeong et al. (2026) document that Epic prioritizes high-potential titles when selecting games for its catalog or promotional events. These two mechanisms select on different characteristics, since one operates through listing costs and the other through expected title performance, but both imply that games listed on Epic's platform are not a random sample. Therefore, this hypothesis highlights the potential issues of comparing Steam-only and dual-platform games without controlling for any pre-existing differences between the two groups.

This hypothesis is tested through the descriptive statistics offered in the results section and the market-level analysis. The descriptive statistics tables that compare

Steam-only and dual-platform games display any differences in prices and sales between the two groups, including in the period before Epic launched its platform. The graphs and event-study regression results in the market-level analysis similarly display any pre-existing differences in prices and sales between the groups. The hypothesis is validated if these two sections of results show statistically significant differences in prices and sales between Steam-only and dual-platform games before the Epic Games Store launched.

3.4 Hypothesis III: Long-Run Pricing Drift

The third and final hypothesis of this study states that pricing and discounting behavior on Steam drifts in the long run in a direction consistent with sustained competitive pressure from Epic. Because the market-level analysis is descriptive, this hypothesis is framed as a directional expectation rather than a causal claim. The expectation is that Epic's growing prominence as the second-largest platform in the market gradually erodes the usage-singlehoming status of Steam's buyer base, which the short-run insulation claim depends on. Steam's dominance in the market relies on retaining its local monopoly over usage-singlehoming buyers, which is not guaranteed indefinitely if multihoming is easily accessible to the buyers (Armstrong, 2006; Caillaud & Jullien, 2003). As for same-market evidence, Jeong et al. (2026) found that Steam consumers became more price-elastic and discounting on Steam became more frequent and deeper within a two-year window around Epic's free-game promotional events. Although this finding appears to contradict the first hypothesis, this effect is driven by repeated promotions as opposed to simply listing the same game, and it is measured over a longer time horizon than the game-level results of this study.

This hypothesis is tested through the market-level analysis results. The event-study regression model displays the annual change in pricing and discounting behavior between Steam-only and dual-platform games, and this model's results for years after Epic's launch provide the best insight into the validity of this claim. Additionally, the change in pricing and discounting behavior is expected to be stronger for games listed on both platforms. However, an important caveat is that the market-level results are not causal, so any changes in pricing and discounting behavior cannot be directly attributed to Epic's long-run competitive pressure.

CHAPTER 4: DATA

4.1 Data Collection

All game data for this project was collected from the IsThereAnyDeal API. This API records every instance of a price change for PC video games across a wide range of storefronts. This pricing data includes the regular price, current price, and discount percentage. Additionally, the API provides metadata for each game such as genre tags, publishers, and release dates.

Monthly CPI data was collected from the Bureau of Labor Statistics to adjust for inflation across the eleven-year sample period. Each price observation was deflated using that month's CPI, with the 2015 average serving as the base year. This produced a real price series alongside the nominal one.

4.2 Data Sample

A sample of 1,826 games was collected from the IsThereAnyDeal API, drawn from the top 1,000 titles in the “Most Popular,” “Most Collected,” and “Most Waitlisted” categories after removing duplicates. This list was then restricted to games released between December 6, 2013, and December 6, 2023, which is five years before and after Epic's entry into the market. Pricing data was collected from December 6, 2013, to December 6, 2024, so that at least one year of pricing data was observed for games released in 2023. Only USD-denominated pricing from the Steam and Epic storefronts was retained. Epic listings were only kept to identify whether and when a game became available on Epic. All outcome variables are measured on Steam.

Several restrictions were then applied to the sample. Free games were excluded because they provide no pricing variation, and games that converted to a free-to-play model during the sample period were dropped after the pricing switch. Downloadable content, prologues, and demos were also removed, since these are not standalone titles. Games listed solely on Epic's platform were dropped because the group was too small to support a separate analysis of Epic-exclusive titles. Games with fewer than ten price observations were then excluded, since these provide too little price variation. Finally, observations with a regular price above \$60 for games released before 2021, or above \$70 for games released afterward, were dropped. These thresholds match the standard retail price for new releases in each period, and observations above these price points likely reflect bundles, premium editions, or listing errors rather than the base price of the game. These restrictions left a final sample of 900 PC video games.

The raw pricing data recorded an observation only when a price changed at a given store. To create a balanced panel with equally weighted observations, this pricing data was expanded into a daily format, with each missing day assigned the last observed price. Each game-store series was also carried forward to the last date the game was sold on that store. This ensured that games without recent price changes remained in the panel through the end of their listing period.

Several variables were constructed from this data. The raw data includes a discount percentage for each observation, which serves as the discount depth outcome and defines a sale indicator equal to one when the discount percentage exceeds zero and zero otherwise. A days-since-release variable was calculated as the difference between the observation date and the game's release date. Metadata for each game provided titles,

release dates, and genre tags. These tags were grouped into fourteen genre categories used in the descriptive statistics.

CHAPTER 5: METHODOLOGY

This section details the methodology used to determine whether Epic’s entry into the market affected pricing or discounting behavior on Steam. The design operates at two levels, market-level and game-level. The market-level analysis studies the long-run relationship between Epic’s launch and outcomes on Steam. All models and graphs for this section either cover the entire data sample period or the entire post-launch period. Due to the length of this timeframe, methods used, and likely non-random selection, these models are only meant to be descriptive and not causal. They provide the evidence relevant to hypothesis II and III. The game-level analysis studies the effect of Epic listing a game on that same title’s pricing and discounting behavior on Steam, and provides the primary evidence for hypothesis I. Although listings occur throughout the post-launch period, the models in that section operate in the short run and are estimated within small time windows around each individual listing date. These estimates support a causal interpretation under the assumption that treated games and their matched controls would have followed parallel paths had the listing never occurred.

5.1 Market-Level Analysis

The market-level analysis is composed of two sets of graphs and an event study regression model. The first set displays the trend in each outcome variable across the entire data sample. All outcome variables are measured on Steam and include real price, sale frequency, and discount depth. Nominal price is excluded from these graphs because inflation over an eleven-year window makes nominal levels difficult to read as a trend,

but the event-study regression below reports both series so the two can be compared directly. To reduced day-to-day noise, each plotted value is the average outcome across all games within a given month. To account for game depreciation over the long sample period, games are split into three age bins covering the first six months after release, six months to two years, and more than two years. The goal is to separate new releases, recent titles, and older games. All graphs contain a dotted vertical line that marks the launch date of the Epic Games Store and separates the pre- and post-launch periods. Within these periods, the graphs show a linear trend fit with a 95% confidence interval and the mean outcome value. Together these graphs describe how each outcome moved before and after Epic launched its platform.

The second set of graphs follows the same construction but compares groups over the post-launch period. Each observation is assigned to one of three groups: Steam-only, dual-platform, and future dual-platform. A game that eventually lists on Epic contributes to the future dual-platform group before its listing date and to the dual-platform group after the listing date. Separating the two groups shows whether games that eventually list on Epic's platform already differed from Steam-only games before the listing date. Each group is plotted with its own trend line and 95% confidence interval. The outcome variables, monthly aggregation, and three game-age bins are identical to those in the first set.

The market-level analysis concludes with an event-study regression covering the entire data sample period. This model is the primary specification of the section and produces its main descriptive results. It is estimated for all four outcomes: nominal price, real price, sale frequency and discount depth. Daily observations are aggregated into

game-month means. The model includes a coefficient for each calendar year, with 2018 as the base year. Because Epic launched on December 6, 2018, that year is almost entirely pre-launch, making it a natural reference point.

An Ever-On-Epic indicator is constructed for this model alone and equals one for any game that appears on both platforms at some point during the sample. Interactions between the year coefficients and Ever-On-Epic separate the year-to-year path between Steam-only games and game that either are or will become dual-platform. To control for game depreciation, a Quarters-Since-Release variable is created that equals the number of quarters an individual game has existed since launch. Because depreciation is fastest in the few years following release and slows down over time, a log transformation of the variable was used as a control in the model. This log specification performed better than its original state and a polynomial alternative in terms of adjusted R^2 . Calendar-month fixed effects control for seasonality, and game fixed effects absorb time-invariant game characteristics. Because Ever-On-Epic does not vary within a game, its level effect is absorbed by the game fixed effects, so the interaction coefficients identify differences in year-to-year movement between the two groups rather than differences in their price levels.

$$Y_{it} = \sum_{k \neq 2018} \beta_k \mathbb{1}[\text{year}_t = k] + \sum_{k \neq 2018} \delta_k (\mathbb{1}[\text{year}_t = k] \times \text{EverOnEpic}_i) \\ + \gamma \log(\text{QSR}_{it}) + \alpha_i + \tau_{m(t)} + \varepsilon_{it}$$

The event-study regression follows the model equation above. Y is the value of the outcome variable for game i during month t . β is the primary coefficient for the Steam-only group and is interpreted as the value of the outcome variable in year k relative to the 2018 base year. δ is the coefficient that is summed with β to obtain the

dual-platform's point estimate for the value of the outcome variable relative to 2018. γ tracks the effect of game depreciation for each game-month combination. Lastly, α and τ are game and month fixed effects, respectively. Standard errors for the model are clustered by game.

In order to interpret the year-relative event-study regression results in a better format, the point estimates for the Steam-only and dual-platform groups are graphed throughout the entire data sample period. The Steam-only estimates are the year coefficients while the dual platform estimates are the sum of the year coefficients and interaction terms between year and Ever-On-Epic. Confidence intervals at the 95% level are reported for both groups to display statistical significance. For the Steam-only group, these are based on the standard error of the year coefficient alone. For the dual-platform group, the point estimate is the sum of the year coefficient and its interaction term. Its standard error is computed from the full variance of that sum, including the covariance between the two coefficients. Given 2018 as the base year, both groups run through zero that year.

5.2 Game-Level Analysis

The goal of this section is to estimate the effect of an Epic listing on that same title's pricing and discounting behavior on Steam within a short time window centered on each game's individual listing date. To test this, three models are used. The first is a two-way fixed effects regression estimated on treated games alone, with no control group. The second is a stacked difference-in-differences design that pairs each treated game with a single matched Steam-only control. The third is a synthetic control difference-in-differences design that builds a weighted Steam-only counterfactual for each treated

game and outcome variable. All three models test nominal price, sale frequency, and discount depth. Nominal price is used over real price since inflation is negligible over windows of at most 360 days. The two difference-in-differences models are the primary specifications of the section. The two-way fixed effects model is reported first as a descriptive benchmark that shows how outcomes move around the listing date without any Steam-only controls, and the difference between those outcomes and the difference-in-differences estimates provides insight on how much movement in the outcomes is attributable to the listing itself.

The first model performed in this section is the two-way fixed effects baseline regression model. The primary goal of this model is to compare the outcome values before and after the listing, and the model tests this through an On-Epic indicator variable that equals 0 before the games lists and 1 after. This On-Epic variable serves as the primary point estimate for the model. To maintain the estimates within a short time frame around the listing date, the following four time windows are tested: 30, 60, 90, and 180 days before and after the listing date. Calendar week fixed effects are included to control for seasonality, and game fixed effects are included to control for game-specific qualities and absorb the baseline price level.

$$Y_{it} = \beta(\text{OnEpic}_{it}) + \alpha_i + \lambda_{w(t)} + \varepsilon_{it}$$

The two-way fixed effects regression model follows the model equation above. Y is the outcome value for game i during day t . OnEpic is the indicator variable that determines if the game is currently listed on Epic's platform, and β is the primary point estimate of the model. β shows the change in the outcome value after the game launches

on Epic. Lastly, α are game fixed effects and λ are fixed effects for the calendar week $w(t)$ containing day t . Standard errors are clustered by game.

The two-way fixed effects regression model examines how each treated game's outcome values change around the listing date, but fails to isolate the effect of listing from movement the game would have experienced regardless. To account for this, the last two models for this section are different variations of difference-in-differences models that match each treated game to a Steam-only counterfactual and compares their change across different time windows. These methods control for seasonality, market-wide shocks, and game age-related depreciation to provide a stronger, less-biased estimate. A causal interpretation of these estimates relies on the assumption that treated games and their matched controls would have followed the same trend if the game never listed on Epic's platform. The two DiD estimators only differ in construction and assignment of control games for each listed title. The stacked design assigns a single matched control for each treated game and outcome variable, selected on a distance score that weights pre-listing trend similarity most heavily alongside outcome level, genre, and release date. The synthetic control design instead restricts the donor pool by genre and window coverage, then fits donor weights to reproduce the treated game's pre-listing path. The event windows match those of the two-way fixed effects model with the exception of the 30-day window, which is dropped because it leaves too few event-weeks on each side of the listing date to establish a pre-trend.

Because games list on Epic's platform at different dates, there is not a consistent calendar date that separates treatment from pre-treatment for all games in the sample. As a result, the two-way fixed effects regression initially used in this section suffers from

bias when treatment effects vary over time (Goodman-Bacon, 2021). To account for this, a stacked difference-in-differences design is adopted (Cengiz et al., 2019; Wing et al., 2024). For each treated game, a self-contained sub-experiment is constructed, centered on that title’s listing date and paired with a Steam-only control. Both the control and treated game are observed over the following time windows centered around the listing date: 60, 90, and 180 days before and after listing. Observations are aggregated to game-weeks and indexed by event-week relative to the listing date. Stacking these sub-experiments into a single panel creates a common estimate while ensuring each treated games is only compared to its assigned control over its own time window.

Each game is assigned a single Steam-only control game, and matching is performed separately for each individual outcome variable. First, a control is only eligible for selection if it shares at least one genre with the treated title and has Steam price and sale data spanning the full window around the game’s listing date. From that list of eligible games, the matched control is selected based on the lowest distance score, which combines four standardization components: the mean squared deviation between the two titles’ demeaned pre-listing outcome paths (weight 1.0), the gap in pre-listing outcome levels (0.3), the degree of genre overlap (0.3), and the difference in release dates (0.1). The matching process values pre-treatment trends over outcome level because the DiD design only requires that the treated and control games move in parallel. Any fixed level difference is absorbed by the sub-experiment fixed effects. Matching is with replacement.

$$Y_{iek} = \beta(Treated_i \times Post_k) + \lambda_{ie} + \tau_{ek} + \varepsilon_{iek}$$

The model above is estimated on the stacked panel. Y is the outcome value for game i , in sub-experiment e , at event-week k . *Treated* indicates the treated game within

each sub-experiment and *Post* indicates event-weeks on or after the listing date. Their coefficient β is the average treatment effect on the treated games. λ are sub-experiment fixed effects by game and τ are sub-experiment fixed effects by event week. The coefficient β is the primary estimate from this model and shows the average change in the outcome for treated games after they were listed on Epic’s platform, controlled by the change in their matched controls. Standard errors are clustered by game.

Although the stacked DiD model provides a control for each treated game and accounts for varying treatment effects over time, a single matched control can leave residual differences in the pre-listing trend, weakening the parallel trends assumption. To pair with the stacked DiD model and account for its limitations, a synthetic control is constructed for each treated title. These synthetic controls are a weighted average of Steam-only games chosen to reproduce the treated game’s pre-treatment outcome trend (Abadie et al., 2010). However, the method departs from the original in two respects to better fit the setting of this study. First, the weights are fit to the demeaned pre-listing series rather than the raw levels. This allows a constant level difference between the treated game and its donors to get absorbed automatically, as opposed to forcing the weights to reconstruct it. This is the constrained-regression form of the synthetic control, which relaxes the original method’s no-intercept restriction (Doudchenko and Imbens, 2016). Second, because the design has many treated titles rather than one, a separate synthetic control is built for each individual game and the resulting treated-versus-synthetic pairs are pooled into a single difference-in-differences (Cavallo et al., 2013). This model is distinct from the synthetic DiD models of Arkhangelsky et al (2021) in that only donor units are reweighted, not time periods.

Similar to the stacked DiD's requirements for control selection, the donor pool consists of Steam-only titles that share at least one genre with the treated title and spans the full time window around the listing date for each treated game and outcome variable. The weights solve a constrained least-squares problem. They minimize the squared distance between the treated title's demeaned pre-treatment trend and the demeaned donor-weights trend, subject to two constraints, non-negativity and weights summing to one. This problem is solved using quadratic programming. The non-negativity and summing constraints also prevent the synthetic control from extrapolating outside the donors' observed range. Because the fit targets the pre-treatment trend as opposed to the level, the level is restored for estimation by adding back the pre-period mean difference to the synthetic series. Fit is then assessed using the pre-treatment root-mean-squared error between the treated and synthetic trends. Weight are estimated separately for each outcome, using that outcome's own donor pool.

$$Y_{dek} = \beta(Treated_d \times Post_k) + \lambda_{de} + \tau_{ek} + \varepsilon_{dek}$$

Each treated title and its synthetic control form a two-unit sub-experiment aligned on event-week. The model above is estimated on the pooled set of these sub-experiments. Y is the outcome value for unit d (the treated game or its synthetic control), in sub-experiment e , at event-week k . *Treated* indicates a treated title and *Post* indicates event-weeks on or after the listing date. Their coefficient β is the average treatment effect on the treated games. λ are sub-experiment fixed effects by treatment status, and τ are sub-experiment fixed effects by event week. As opposed to the stacked DiD, standard errors are clustered by sub-experiment because the synthetic control in each pair is a constructed series rather than an independent title.

CHAPTER 6: RESULTS

6.1 Descriptive Statistics

In addition to the market and game-level results, descriptive tables comparing Steam-only and dual-platform game characteristics were created. These tables are meant to provide insight into any pre-existing differences between the groups and justify the use of game fixed effects for controlling omitted variable bias. Table 6.1 shows the average values of each outcome variable for Steam-only and dual-platform games across the pre-launch, post-launch, and full sample periods. Steam-only contains games that only list on Steam throughout the entire study period, and dual-platform defines all games sold on Steam that are either also currently sold on the Epic Games Store or will be in the future. This explains why some games are labeled dual-platform before Epic ever launches their platform. The average estimates are calculated by splitting the data into the relevant sample period, calculating the average outcome value for each game, and then averaging those game-level values across all games in that period. This gives each game equal weight regardless of how many observations it contributes. Because games observed for only a few months are disproportionately recent releases, and recent releases carry higher prices with shallower, less frequent discounts, this weighting exerts upward pressure on average prices and downward pressure on sale frequency and discount depth. The t-tests reported in the difference column are conducted on these game-level averages.

Table 6.1 Outcome Variable Comparison by Group

Variable	Steam-Only	Dual-Platform	Difference
Pre-Epic Period (before Dec 2018)			
Avg Price (USD)	23.76	28.20	4.45***
Avg Real Price (USD)	22.90	27.19	4.29***
Sale Frequency	0.12	0.15	0.02**
Discount Depth (%)	5.57	5.65	0.08
N (games)	335	174	
Post-Epic Period (after Dec 2018)			
Avg Price (USD)	24.05	27.71	3.66***
Avg Real Price (USD)	19.95	22.63	2.68***
Sale Frequency	0.19	0.22	0.03***
Discount Depth (%)	10.24	11.85	1.62***
N (games)	544	356	
Full Sample Period			
Avg Price (USD)	24.56	28.41	3.85***
Avg Real Price (USD)	20.85	23.66	2.82***
Sale Frequency	0.18	0.21	0.03***
Discount Depth (%)	9.24	10.83	1.58***
N (games)	544	356	

Table 6.1 displays statistically significant differences in every outcome variable between Steam-only and dual-platform averages across every period. The only exception is discount depth in the pre-launch period. The differences in average nominal and real prices are large in each period. Before the Epic Games Store launched, the nominal price difference between Steam-only and future dual-platform games was \$4.45, which falls only to \$4.29 when adjusting for inflation. The difference in sale frequency in this period is 2 percentage points despite reaching statistical significance. These results indicate a substantial pre-existing difference in the price and discounting profile of the two groups.

The differences remain significant across every outcome in the period after Epic launched, but the price gap narrows to \$3.66 in nominal terms and \$2.68 in real terms. In nominal terms this narrowing comes from Steam-only prices rising slightly while dual-platform prices fall. In real terms both groups fall, and the gap narrows because dual-platform prices fall further. Sale frequency increases for both groups, with the gap widening only to 3 pp, and discount depth rises for both groups with dual-platform games holding a slightly larger average value. Two caveats apply to these period comparisons. First, they are not computed on a balanced panel, since roughly 40% of the games in the post-launch sample had not yet released before December 2018. Therefore, the pre/post difference reflects both changes in the prices of existing games and the entry of new titles into the sample. Second, only the post-launch period contains games more than five years old, which puts downward pressure on average prices and upward pressure on discounting in that period for both groups. Both features distort the pre/post comparison more than the pre-existing gap between the groups, and both point to the need for a game age control in the regression models that follow.

Table 6.2 Release Year Distribution by Group

Release Year	Steam-Only (%)	Dual-Platform (%)	Difference
2013	0.6%	0%	-0.6
2014	11.4%	6.5%	-4.9**
2015	12.5%	8.4%	-4.1*
2016	11.6%	9.6%	-2
2017	10.7%	12.1%	1.4
2018	11.4%	10.4%	-1
2019	9.2%	9%	-0.2
2020	6.6%	9.8%	3.2

Release Year	Steam-Only (%)	Dual-Platform (%)	Difference
2021	7.7%	11.8%	4.1*
2022	9%	10.7%	1.7
2023	9.4%	11.8%	2.4

Continuing with the descriptive tables, Table 6.2 shows the distribution of release years across Steam-only and dual-platform games. Most release years show statistically insignificant differences between the two groups, with three exceptions. The Steam-only sample has a significantly larger share of games released in 2014 and 2015, and the dual-platform sample has a significantly larger share released in 2021. This skew is expected, since Epic's platform launched five years after the start of the data collection window, and Epic likely targeted newer, higher-profile titles as a promotional tool to attract users. The result is that the Steam-only sample contains somewhat older games than the dual-platform group. Because games depreciate, this creates a downward bias on Steam-only prices and an upward bias on Steam-only sale frequency and discount depth. This sharpens the reading of Table 6.1, which shows Steam-only games with lower prices but also lower sale frequency and discount depth in both periods. Once game age is accounted for, the age-adjusted price difference is likely smaller and the age-adjusted sale frequency and discount depth differences likely larger than Table 6.1 indicates.

Table 6.3 Genre Distribution by Group

Genre	Steam-Only (%)	Dual-Platform (%)	Difference
Adventure	21.3%	31.2%	9.9***
OpenWorld	16.9%	21.9%	5*
RPG	30.5%	25.6%	-5
Narrative	15.4%	20.2%	4.8*
Multiplayer	20.2%	17.7%	-2.5

Genre	Steam-Only (%)	Dual-Platform (%)	Difference
Survival	11.2%	13.5%	2.3
Platformer	9.6%	11.5%	2
Shooter	11%	9.3%	-1.8
Horror	8.1%	9.8%	1.7
Strategy	12.9%	14.6%	1.7
Puzzle	5.3%	6.7%	1.4
Action	32.2%	30.9%	-1.3
Simulation	13.1%	13.5%	0.4
Roguelike	7.2%	7%	-0.1

Lastly, Table 6.3 displays the share of genres across Steam-only and dual-platform games. An important clarification is that a game can feature multiple genres, resulting in the columns not summing to 100%. As with the release year distribution, most genres show statistically insignificant differences between the two groups. The only exceptions are adventure, open world, and narrative, each of which appears at a higher rate among dual-platform games. The genre and release year imbalances documented in Tables 6.2 and 6.3, along with the outcome differences in Table 6.1, motivate the empirical strategy used throughout the study. Game fixed effects are included in every regression model, which allows the models to control for time-invariant game characteristics that would otherwise bias the results. The outcome and release date differences additionally motivate the time since release control included in the regression models.

6.2 Market-Level Analysis

This section examines the market-level analysis results. The results here are not meant to be read as causal estimates of the impact of Epic's entry, but they do provide a

descriptive outlook on the PC game market and how Epic’s entry may have affected Steam in the long run. The section presents two sets of figures, with the first covering the full sample period and the second covering the post-launch period only. Each figure plots one outcome variable over time, with the post-launch figures separating games into Steam-only, future dual-platform, and current dual-platform groups. The section then presents an event-study regression model that estimates a coefficient for each year, along with an interaction between each year and the dual-platform indicator. These results are also visualized through time-series figures displaying each group's point estimates and confidence intervals. Game depreciation is controlled through the log of quarters since release (QSR), seasonality through month-of-year fixed effects, and time-invariant game characteristics through game fixed effects. Although the results are not causal, these controls provide a more accurate description of how Steam prices and discounting shifted across the sample period than the raw figures alone.

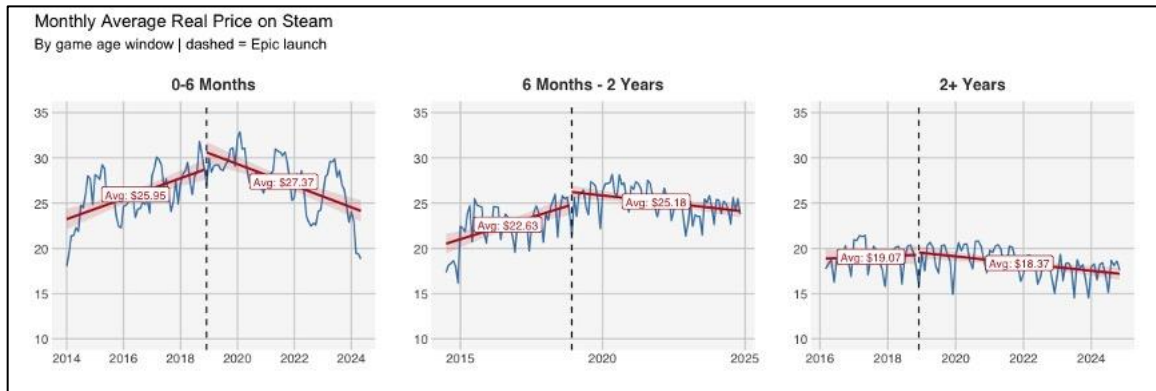


Figure 6.1 Monthly Average Real Price on Steam

The first set of figures shows outcome variable trends across the full sample period and is split into three game-age windows (0–6 months, 6 months–2 years, and 2+ years). Figure 6.1 shows real price over the sample period and indicates distinct trends between the pre- and post-launch periods. Within the 0–6 month window, the average

real price of games on Steam is ~\$1.40 higher in the post-launch period, but the trends differ sharply. Steam prices were on an upward trajectory before Epic's launch and turned to a negative slope afterward. In the 6 months–2 years window the average price increases by more (~\$2.50), but the post-launch trend is not as negatively sloped. The 2+ years window has very similar averages in both periods, with prices falling ~\$0.70, and the trend stays relatively flat throughout. Trend lines for the first two windows rise before Epic's launch and turn negative afterward, while the oldest window stays close to flat in both periods.

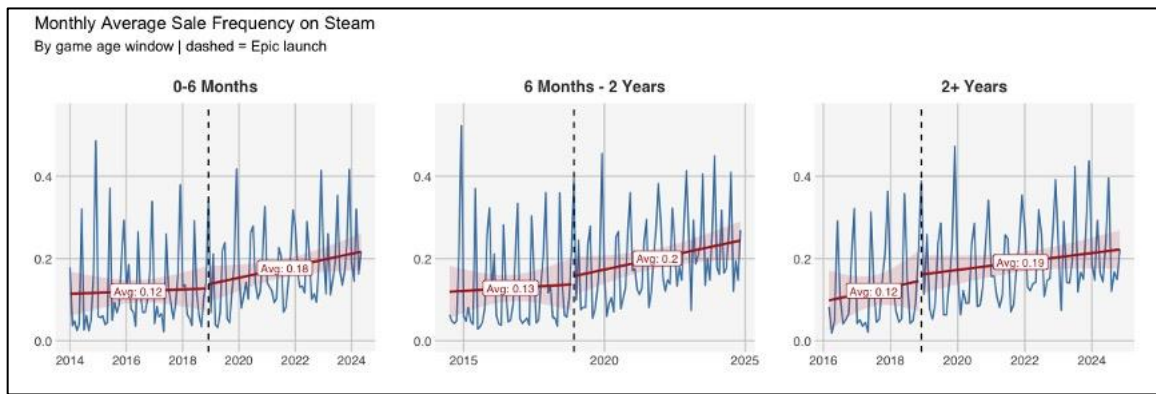


Figure 6.2 Monthly Average Sale Frequency on Steam

Figure 6.2 shows the changes in sale frequency on Steam before and after Epic launches their platform. All three game age windows have relatively similar results in terms of the difference in averages before and after, but the trend lines differ slightly. The average monthly sale frequency increases by 6-7 pp after Epic's entry into the market for all three game age windows. As for the difference in trends, the 0-6 months and 6 months – 2 years game-age windows have relatively flat change in sale frequency before Epic's entry and a positive slope after. The 2+ years window differs in that sale frequency was already trending upward before Epic's launch, but the post-launch trend is steeper still.

This indicates that sale frequency for games of all different release dates grew after Epic's launch.

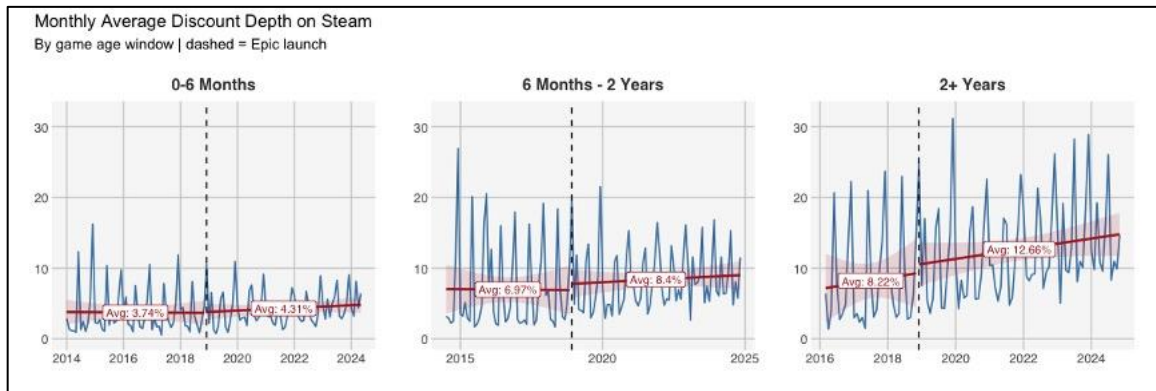


Figure 6.3 Monthly Average Discount Depth on Steam

Lastly, Figure 6.3 examines changes to discount depth before and after Epic launched its platform. All three game-age windows show similar patterns, but both the difference in averages and the change in slope strengthen as games get older. For the newest games, discount depth increases by only ~0.5 pp in the post-launch period, and the trend line stays close to flat in both periods. For the 6 months–2 years window, average discount depth increases ~1.4 pp and the slope shifts from flat to positive. The 2+ years window shows the largest increase at ~4.4 pp, with the trend line becoming increasingly positive after Epic's launch. Overall, Epic's entry coincides with an increase in the average monthly discount depth of games on Steam, but this pattern is concentrated among older titles.

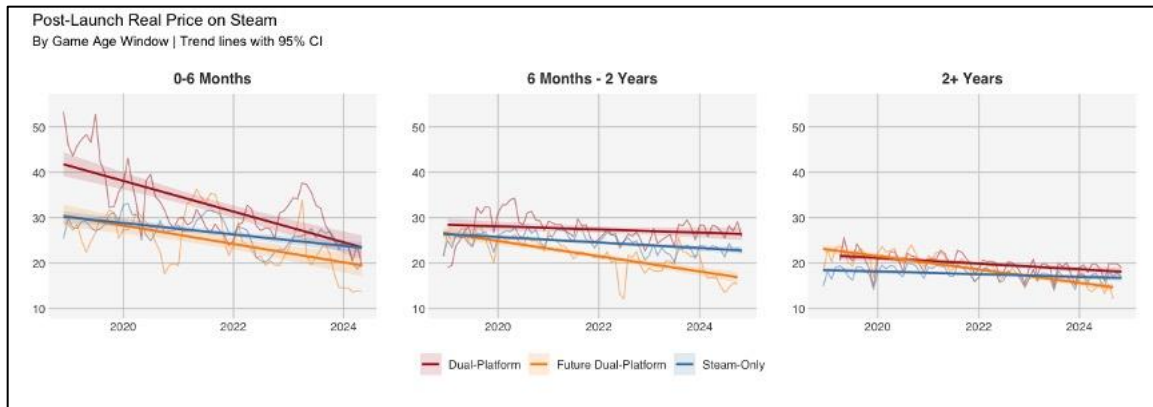


Figure 6.4 Post-Epic Launch Real Prices on Steam Group Comparison

The second set of figures presents post-launch group comparisons across the same three game-age windows. The three groups are games listed only on Steam (Steam-only), games sold on both platforms (dual-platform), and games that will sell on both platforms in the future (future dual-platform). Figure 6.4 shows real price trends and reveals convergence across the three groups as games age. The 0–6 month window has the largest spread, with dual-platform games starting roughly \$11 to \$12 above the other two groups before falling to the Steam-only level by the end of the period. Steam-only and future dual-platform games begin at the same price, but future dual-platform games follow a slightly steeper downward trend. All three groups have negative trend lines through the post-launch period, with dual-platform declining most steeply. For the 6 months to 2 years window, all three groups start at a similar price level, but Steam-only and dual-platform only have a slight decline in prices, while future dual-platform games have a much steeper fall in prices. The 2+ year window shows all three groups declining. Steam-only and dual-platform decline gently and end at similar levels, while future dual-platform games fall much further. Together these figures indicate that post-launch price differences between the groups exist primarily among newer games.

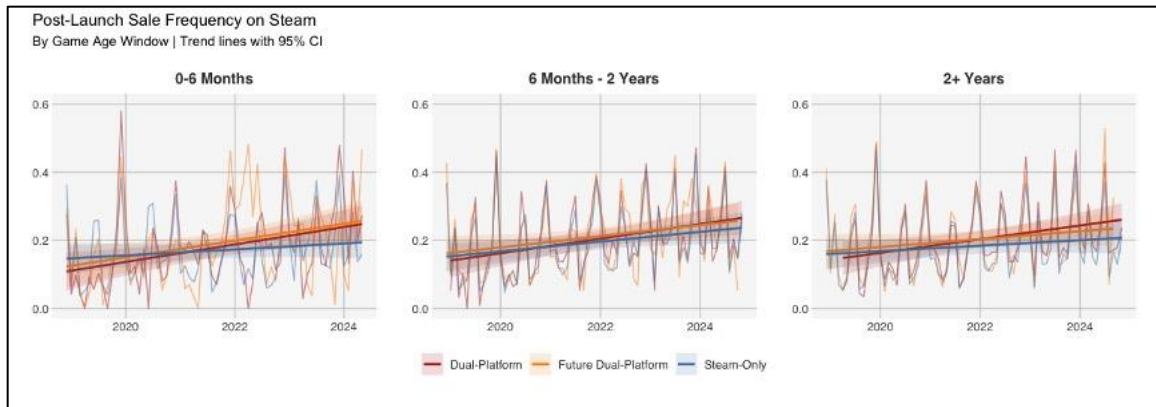


Figure 6.5 Post-Epic Launch Sale Frequency on Steam Group Comparison

The next set of group comparison figures (Figure 6.5) tracks changes to sale frequency across the post-launch period. In contrast to the price results, sale frequency across the three groups shows similar values and slopes in all three game-age windows. In the 0–6 month window, Steam-only games follow a mildly positive trend while dual-platform and future dual-platform games rise somewhat more steeply. Despite this difference, the average monthly sale frequency for all three groups falls into a narrow band between roughly 0.10 and 0.25. In the 6 months to 2 years window, all three groups have nearly identical trend lines and values, and in the 2+ year window dual-platform and future dual-platform games show a slightly steeper positive slope at near-identical levels. Overall, the three groups did not differ meaningfully in sale frequency across the post-launch period.

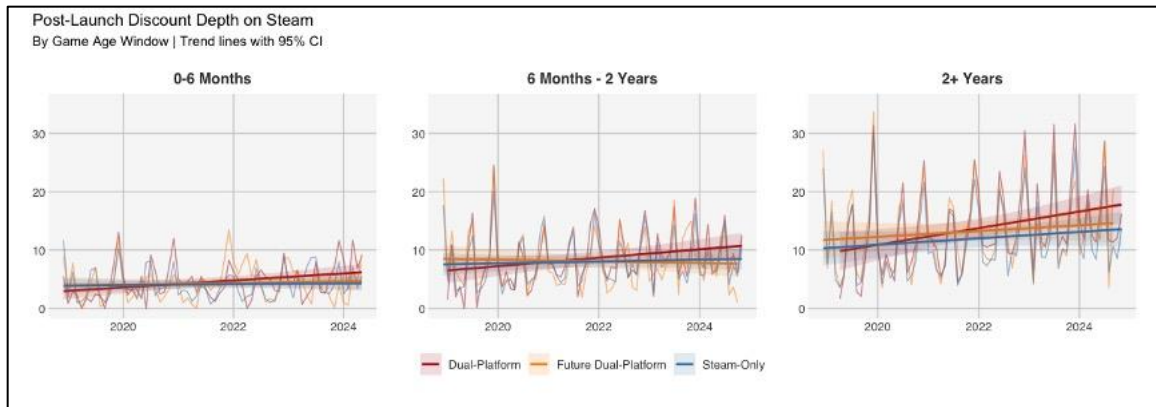


Figure 6.6 Post-Epic Launch Discount Depth on Steam Group Comparison

The last set of group comparison figures (Figure 6.6) examines changes in discount depth over the post-launch period. Unlike the sale frequency results, discount depth does differ modestly across the three groups in both average value and slope, though the difference only exists for older games. Newer games in the 0-6 month window experienced almost no change in discount depth across all three groups, with the groups tracking one another closely. The 6 month to 2 year window begins to show separation, with the dual-platform group having a positive slope against flatter trend lines for the other two groups, but all three groups remain within 7% to 11%. In the 2+ year window, Steam-only and future dual-platform games share a small positive trend and similar average discount depth, while current dual-platform games follow a steeper trend and have a higher average. Overall, the three groups do not diverge greatly in discount depth across the post-launch period, but dual-platform games hold slightly larger average values and steeper positive trends.

Table 6.4 Market-Level Regression Results

	Nominal Price (USD)	Real Price (USD)	Sale Freq. (share)	Discount Depth (pp)
2014	0.123	0.083	0.054**	2.557**
	(1.057)	(1.056)	(0.019)	(0.803)

	Nominal Price (USD)	Real Price (USD)	Sale Freq. (share)	Discount Depth (pp)
2015	1.121	1.294	0.004	1.384*
	(0.816)	(0.818)	(0.010)	(0.603)
2016	0.480	0.770	-0.006	0.742*
	(0.540)	(0.537)	(0.006)	(0.354)
2017	0.016	0.235	-0.008	-0.072
	(0.278)	(0.272)	(0.004)	(0.222)
2019	-0.116	-0.208	0.021***	0.912**
	(0.216)	(0.207)	(0.005)	(0.297)
2020	-0.302	-0.393	0.011	0.081
	(0.313)	(0.302)	(0.006)	(0.351)
2021	-0.665	-1.291***	0.016*	0.231
	(0.396)	(0.383)	(0.006)	(0.404)
2022	-0.966*	-2.758***	0.038***	1.468**
	(0.438)	(0.425)	(0.008)	(0.508)
2023	-1.224**	-3.544***	0.059***	2.558***
	(0.466)	(0.458)	(0.009)	(0.593)
2024	-1.212*	-3.849***	0.066***	2.910***
	(0.517)	(0.508)	(0.009)	(0.633)
2014 x Epic	5.190*	5.426*	-0.044	-1.163
	(2.603)	(2.590)	(0.026)	(1.042)
2015 x Epic	3.699*	4.092*	-0.018	-0.609
	(1.756)	(1.763)	(0.014)	(0.773)
2016 x Epic	1.643	1.945	-0.001	-0.240
	(1.009)	(1.014)	(0.009)	(0.517)
2017 x Epic	0.858	0.953	0.004	-0.266
	(0.553)	(0.545)	(0.009)	(0.327)
2019 x Epic	-0.462	-0.483	-0.003	0.466
	(0.341)	(0.328)	(0.007)	(0.394)
2020 x Epic	-0.787	-0.835	-0.003	0.319

	Nominal Price (USD)	Real Price (USD)	Sale Freq. (share)	Discount Depth (pp)
	(0.508)	(0.490)	(0.008)	(0.472)
2021 x Epic	-0.726	-0.897	0.000	0.745
	(0.598)	(0.585)	(0.008)	(0.503)
2022 x Epic	-1.506*	-1.721*	0.016	2.026**
	(0.686)	(0.667)	(0.010)	(0.677)
2023 x Epic	-1.657*	-1.904**	0.012	2.316**
	(0.709)	(0.692)	(0.011)	(0.783)
2024 x Epic	-1.922**	-2.144**	0.021*	3.408***
	(0.714)	(0.703)	(0.010)	(0.756)
Log(QSR)	-1.219***	-1.682***	0.008	2.994***
	(0.252)	(0.247)	(0.004)	(0.207)
Num Obs	69654	69654	69654	69654
R2 Within	0.128	0.309	0.037	0.128

* p < 0.05, ** p < 0.01, *** p < 0.001

All models include game and month-of-year fixed effects. Reference year 2018. SEs clustered by game.

The market-level analysis concludes with an event-study regression model that estimates annual coefficients across the full sample period, with 2018 as the reference year. The model controls for time since release, seasonality, and time-invariant game characteristics, making its results carry more weight than the raw figures. However, the results remain descriptive rather than causal. Two features of Table 6.4 require care in interpretation. First, because the model includes game fixed effects, any permanent level difference between the Steam-only and dual-platform groups is mechanically absorbed and cannot be examined from these coefficients. Second, the year coefficients describe the Steam-only group's movement relative to 2018, while the interaction terms describe how much further or less far the dual-platform group moved over the same time frame.

Significance in the year coefficients is concentrated in the discounting outcomes. Three of the ten year coefficients reach significance for nominal price and four of ten for real price, compared to six of ten for sale frequency and seven of ten for discount depth. For nominal price, significance appears from 2022 through 2024, but it begins in 2021 for real price. The interaction terms are large and positive in 2014 and 2015, at roughly \$4 to \$5, and significantly negative from 2022 through 2024 for both price outcomes. The interaction terms decline over the pre-launch period, from +5.19 in 2014 to +3.70, +1.64, and +0.86 in the following years. Because these coefficients measure differential change rather than level differences, this pattern indicates that dual-platform prices were falling faster than Steam-only prices well before Epic launched. The pre-launch coefficients fall much faster than the post-launch coefficients, arguing against the post-launch trend occurring from Epic's entry.

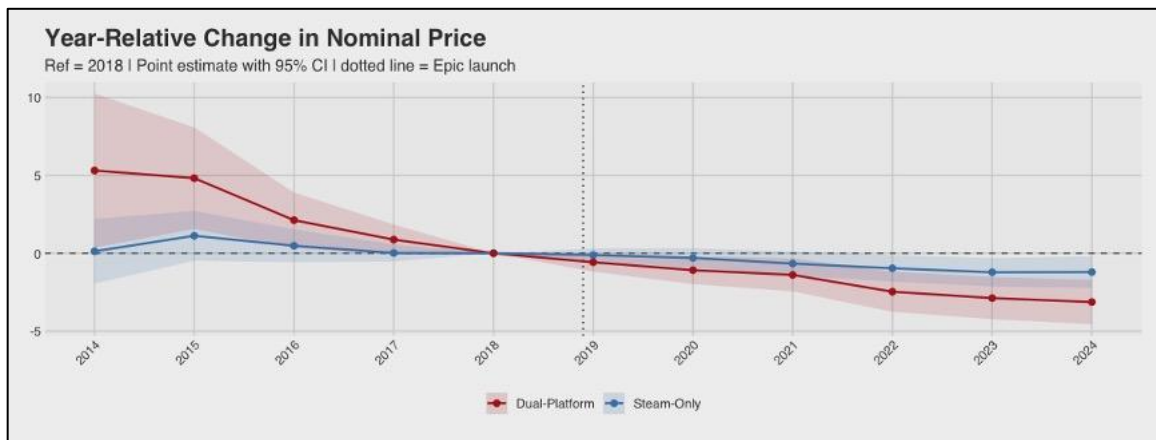


Figure 6.7 Market-Level Nominal Price Regression Graph

Figure 6.7 plots the year-relative change in nominal price for both groups, normalized to 2018. The two lines pass through zero in 2018 by construction, so the vertical distance between them in any year represents the differential change relative to that base year rather than a difference in price levels. Given this, the figure shows that

dual-platform prices in 2014 and 2015 were significantly higher than their 2018 level compared to Steam-only prices, and that this differential shrinks steadily through 2018 and continues to shrink post-launch. Steam-only prices remain relatively stable across the eleven years of the sample. Based on the confidence intervals, dual-platform prices are significantly above their 2018 level, relative to Steam-only, in 2014 and 2015 only. Both groups show significantly lower prices from 2022 through 2024 relative to 2018.

Although the confidence intervals overlap, dual-platform prices fall by more. Figure 6.8 shows the same structure for real price, with larger declines for both groups from 2022 through 2024 and an additional significant year in 2021. Both nominal and real prices fall for both groups several years after Epic's entry, relative to 2018. The two groups were already converging from 2014 through 2018, and at a faster rate than they converged afterward.

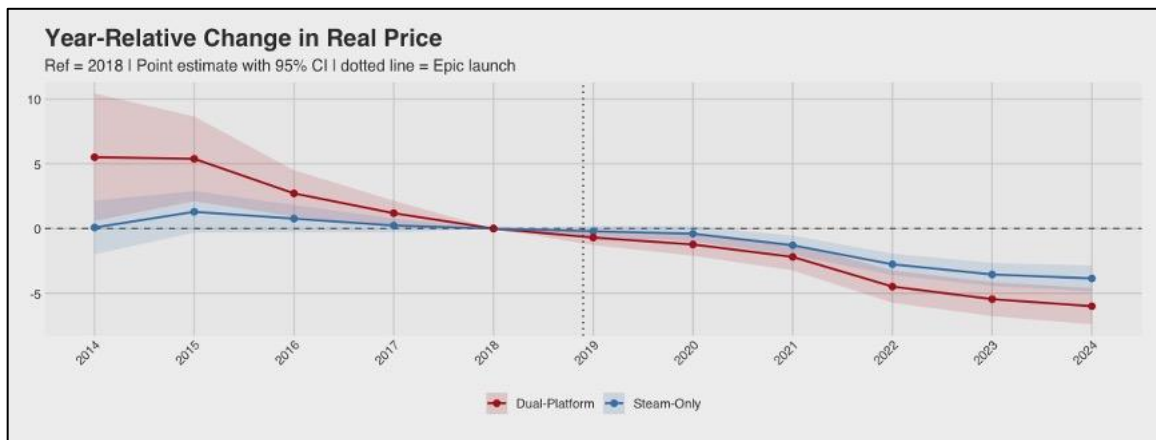


Figure 6.8 Market-Level Real Price Regression Graph

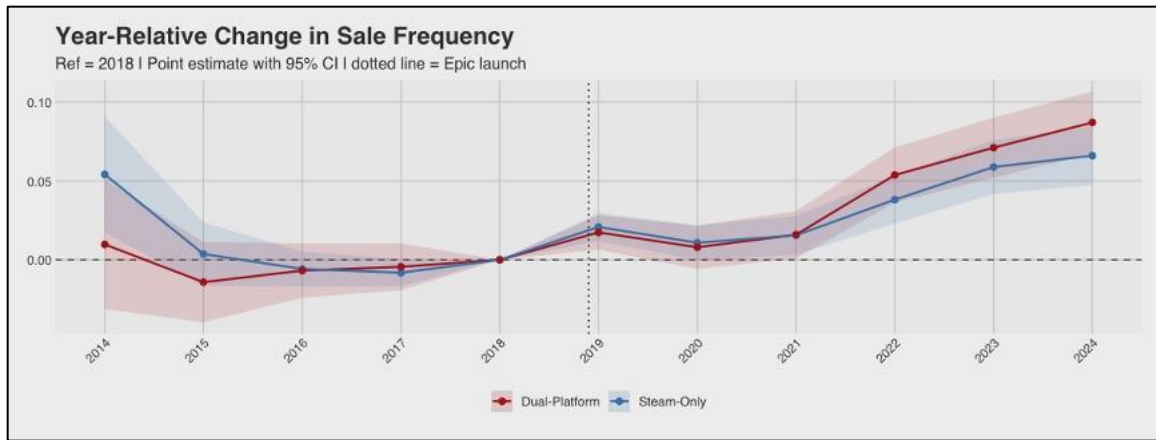


Figure 6.9 Market-Level Sale Frequency Regression Graph

As for the findings on discounting behavior, both sale frequency and discount depth rise sharply toward the end of the sample period. Figure 6.9 plots the year-relative change in sale frequency, again normalized to 2018. Just like the price figures, the distance between the lines measures differential movement rather than a level gap, so the higher Steam-only position in 2014 indicates that Steam-only sale frequency fell further between 2014 and 2018, not necessarily that Steam-only games discounted more often than dual-platform games. This is consistent with the level differences reported in Table 6.1, since the two figures measure different quantities. Over the middle of the sample the two groups' estimates converge almost completely. Only one of the ten sale frequency interaction terms reaches significance, in 2024, so the two groups' discounting frequency moved almost identically across the sample and their paths separate only in the final year. Both groups show significant increases relative to 2018 from 2022 through 2024, with earlier significant years for Steam-only in 2019 and 2021. Sale frequency rises by roughly 6.6 pp for Steam-only games and 8.7 pp for dual-platform games by 2024. Figure 6.10 shows the corresponding results for discount depth, where both groups are significantly higher from 2022 through 2024 and dual-platform games reach a

substantially larger increase by 2024 at roughly 6.3 pp against 2.9. Unlike sale frequency, the discount depth interaction terms are significant in 2022, 2023, and 2024, so this is the one discounting outcome where the two groups genuinely diverge.

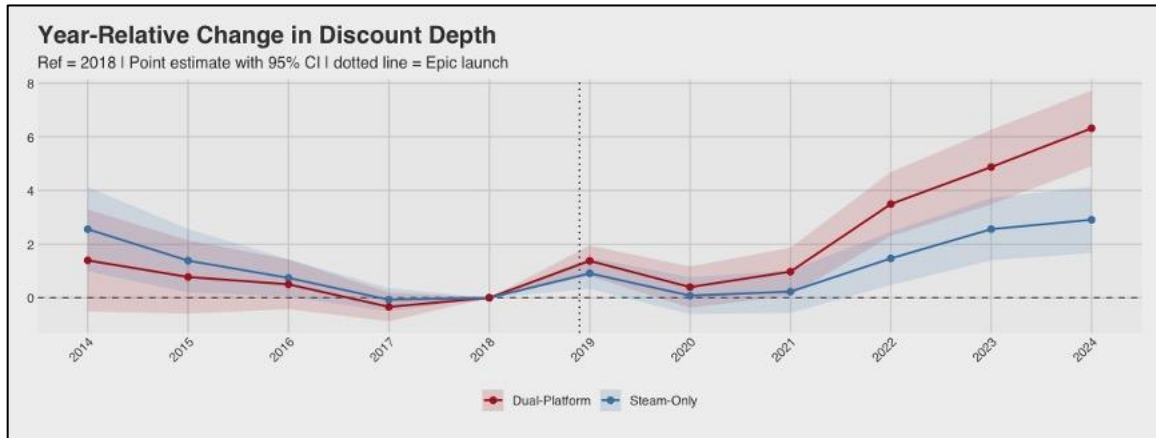


Figure 6.10 Market-Level Discount Depth Regression Graph

Collectively, these market-level findings show that nominal and real prices fell while sale frequency and discount depth rose for both Steam-only and dual-platform games, with the movement concentrated three to six years after Epic launched its platform. These changes are somewhat stronger for dual-platform games, most clearly so for discount depth, where the group differential is itself statistically significant from 2022 onward. Two qualifications limit how much of this can be attributed to Epic. First, the divergence between the nominal and real price paths after 2021 is largely mechanical. Nominal prices fall by ~\$1.21 by 2024 while real prices fall by \$3.85, so roughly two thirds of the real decline reflects the deflator rather than movement in observed prices. Publishers largely held nominal prices fixed and inflation eroded the real value of those unchanged prices, so the real decline reflects that inaction rather than active price cutting. Second, the two groups were already converging from 2014 through 2018, and at a faster annual rate than they converged after Epic's entry, which weakens any account in which

Epic's arrival produced the convergence. What these results do establish is that the two groups were already on different pricing paths before Epic existed, which is consistent with Table 6.1's level differences and confirms that Epic's listings were not randomly assigned across titles.

6.3 Game-Level Analysis

This section examines the results of the game-level analysis performed in this paper. The section begins with the two-way fixed effects regression model that tests the impact on pricing and discounting behavior on Steam when the same title lists on Epic. These models test nominal price, sale frequency and discount depth in the following four time windows around the listing date: 30, 60, 90, and 180 days before and after the listing. In addition to this baseline model, two difference-in-differences models are constructed in this section that also test the impact of the listing on pricing and discounting behavior on Steam, while also utilizing Steam-only games as controls. The two models differ in their control selection and creation. The first model is a stacked DiD with one Steam-only matched control and the second model is a synthetic control DiD that generates controls based on a weighted average of Steam-only games. Additionally, these DiD models differ from the two-way fixed effects model by excluding the 30 day time window due to the small number of event weeks. The DiD models provide more causal results on the impact of Epic's listing, while the baseline model primarily serves as a robustness check.

Table 6.5 Game-Level Two-Way Fixed Effects Model Results on Price

	30d	60d	90d	180d
On Epic (post-entry)	-0.236	-0.183	0.039	0.180
	(0.447)	(0.373)	(0.341)	(0.263)

	30d	60d	90d	180d
Num Obs	17817	34933	51731	101065
R2 Within	0.000	0.000	0.000	0.000

* p < 0.05, ** p < 0.01, *** p < 0.001

Table 6.6 Game-Level Two-Way Fixed Effects Model Results on Sale Frequency

	30d	60d	90d	180d
On Epic (post-entry)	0.023	-0.003	-0.002	-0.009
	(0.025)	(0.021)	(0.019)	(0.014)
Num Obs	17817	34933	51731	101065
R2 Within	0.000	0.000	0.000	0.000

* p < 0.05, ** p < 0.01, *** p < 0.001

Table 6.7 Game-Level Two-Way Fixed Effects Model Results on Discount Depth

	30d	60d	90d	180d
On Epic (post-entry)	0.201	0.137	-0.185	-0.157
	(1.406)	(1.118)	(0.996)	(0.718)
Num Obs	17817	34933	51731	101065
R2 Within	0.000	0.000	0.000	0.000

* p < 0.05, ** p < 0.01, *** p < 0.001

For the two-way fixed effects baseline model, all three outcome variables tested return null results, indicating that a game listing on Epic had no impact on that same title's pricing or discount behavior on Steam in the short run. Table 6.5 displays the results for nominal price, and fails to provide statistically significant or consistent results. Prices appear to fall slightly within the 30 and 60 day windows, but rise after 90 to 180 days. As for the sale frequency results, Table 6.6 also contains no statistically significant results. Within 30 days of the listing, sale frequency has a small increase, but this effect

flips and becomes much smaller for the longer time windows. Lastly, Table 6.7 presents the discount depth results and contains similar findings to the other models. The effect of the listing on discount depth is statistically insignificant for all four time windows, and the direction of the change differs between these time windows. Collectively, the two-way fixed effects baseline model results indicate that a game listing on Epic has no statistically significant effect on prices, sale frequency, or discount depth for the same title on Steam within the short run. Although this model's results are not the strongest or most causal, they can provide insight when compared to the DiD findings.

Before interpreting the DiD results, the pre-treatment fit these results rely on is assessed. Each time window and model combination estimates a treated-versus-control coefficient for every event week, relative to the week prior to listing. These coefficients measure whether a treated game and its control were already diverging before the title listed on Epic's platform. Table 6.8 reports a joint Wald test of the null hypothesis that all pre-treatment coefficients equal zero, estimated separately for each outcome, time window, and DiD model combination. To match the corresponding model, standard errors are clustered by game. A high p-value means the test fails to reject the null of no pre-treatment divergence, while a low p-value indicates the paths were already diverging before the listing. This test speaks only to the pre-treatment period and does not establish that parallel trends would have held after the listing date. Seventeen of the eighteen tests fail to reject the null. The exception is the 180 day synthetic control model on price, with a p-value of 0.008. As a result, that specification cannot support a causal reading.

Table 6.8 Game-Level Difference-in-Differences Model Pre-Treatment Fit

Outcome	Window (days)	Joint pre-trend test (p-value)	
		Stacked DiD	Synthetic Control DiD
Price (USD)	60	0.212	0.402
	90	0.278	0.327
	180	0.161	0.008
Sale Frequency	60	0.811	0.101
	90	0.570	0.343
	180	0.567	0.057
Discount Depth (%)	60	0.535	0.368
	90	0.415	0.656
	180	0.357	0.165
H0: all pre-period event-study coefficients = 0 jointly (Wald chi-square, reference week = -1). High p-values support parallel trends. 17 of 18 tests pass.			

The stacked and synthetic control results are explored below by outcome variable. Table 6.9 displays the price results for both DiD models and contains the first statistically significant estimates in the game-level analysis. Both models find a significant increase in the Steam price of a title within 90 days of that same title listing on Epic, at \$0.48 and \$0.53 respectively, or roughly 2% of the average game price. However, two caveats are worth noting. First, the 60 day stacked estimate of \$0.50 is larger than the significant 90 day estimate and differs only in precision, so the pattern across windows is better described as a consistently small positive point estimate that crosses conventional significance thresholds in the 90 day window rather than an effect that appears and then disappears. Second, these estimates come from eighteen model, window, and outcome combinations, and the windows are nested rather than independent. The price result being statistically significant in the same window for both models strengthens the finding considerably. The synthetic control estimate for the 180 day window cannot be

interpreted given the failed pre-trend test in Table 6.8, but the stacked estimate at that window can, and its smaller magnitude of \$0.14 suggests the effect does not persist. The direction of this finding deserves emphasis. Epic's listing did not exert downward pressure on Steam prices for the same title, and if anything pushed prices upward.

Table 6.9 Game-Level Difference-in-Differences Model Results on Price

	Stacked DiD			Synthetic Control DiD		
	60d	90d	180d	60d	90d	180d
ATT (treated x post)	0.499	0.475*	0.138	0.348	0.530*	0.233
	(0.277)	(0.231)	(0.199)	(0.248)	(0.207)	(0.196)
Num Obs	8596	12110	22986	8586	12104	22974
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$						

Examining the estimates for sale frequency from these two DiD models, Table 6.10 shows that a game listing on Epic has a statistically insignificant effect on the same title's sale frequency on Steam. Both models in all three time windows fail to find any significant results. Along with this, the coefficients flip signs throughout the time windows and remain very small. Therefore, it is safe to say that sale frequency remains unaffected on Steam.

Table 6.10 Game-Level Difference-in-Differences Model Results on Sale Frequency

	Stacked DiD			Synthetic Control DiD		
	60d	90d	180d	60d	90d	180d
ATT (treated x post)	-0.006	-0.002	0.005	0.009	0.009	0.018
	(0.011)	(0.009)	(0.008)	(0.012)	(0.009)	(0.010)
Num Obs	8596	12110	22986	8586	12104	22974
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$						

Lastly, the two DiD models' results for discount depth are statistically insignificant with one exception. Within 180 days of a game listing on Epic's platform, the synthetic control model estimates that the discount depth of that same title on Steam rises by roughly 1.2 pp. The stacked estimate at the same window is 0.356 with a standard error of 0.472, so the two confidence intervals overlap substantially. All other windows show no change in discount depth, with several coefficients flipping signs. Given the synthetic control model's preferred standing in this study and its clean pre-trend test at this window, this estimate is worth taking seriously. However, it is a single significant result from one model at one window within a group of eighteen estimates, and it should be weighted accordingly.

Table 6.11 Game-Level Difference-in-Differences Model Results on Discount Depth

	Stacked DiD			Synthetic Control DiD		
	60d	90d	180d	60d	90d	180d
ATT (treated x post)	-0.119	-0.048	0.356	0.500	0.349	1.186**
	(0.605)	(0.549)	(0.472)	(0.560)	(0.476)	(0.374)
Num Obs	8596	12110	22986	8586	12104	22974
* p < 0.05, ** p < 0.01, *** p < 0.001						

Overall, the DiD results point mostly to an insignificant effect of Epic listing on pricing and discounting behavior for the same title on Steam, with two exceptions. First, the Steam price of a game rises ~\$0.50 within 90 days of that title listing on Epic. Both DiD models detect this effect, but the 180-day stacked estimate suggests it does not persist. Second, discount depth rises ~1.2 percentage points within 180 days according to the synthetic control model alone. Because the price outcome is the observed current price on Steam, and sale frequency and discount depth show no significant change within

the same 90-day window, the price increase appears to operate through a higher list price rather than through reduced discounting. These null results are not in tension with the market-level findings. The game-level design tests windows of at most 180 days around an individual listing, while the market-level shifts in price and discounting appear three to six years after Epic's entry. Taken together, the two analyses suggest that Epic's entry did not trigger a title-level pricing response on Steam, and that whatever market-wide adjustment occurred did so gradually and through channels other than the individual listing decision.

CHAPTER 7: DISCUSSION

7.1 Summary of Findings

At the market level, the findings support two claims. First, the descriptive tables, along with the price gap visible among the newest games in the group comparison figures, indicate substantial differences in characteristics between games listed only on Steam and games listed on both platforms. These differences are present in the pre-launch period and risk contaminating any market-level comparison. Second, the findings show downward movement in prices and rising sale frequency and discount depth over the sample period. To account for sampling and depreciation bias, the market-level regression model controls for game age, seasonality, and time-invariant game characteristics. This model produces significant estimates of falling prices and rising sale frequency and discount depth in the later years of the study. The differential movement between groups is statistically significant for both price outcomes and for discount depth, but not for sale frequency, where only the final year separates them. These movements are concentrated three to six years after Epic's entry, and two features of the data limit how much can be attributed to it. The two groups were already converging before Epic launched, at a faster annual rate than they converged afterward, and roughly two thirds of the real price decline reflects the deflator rather than movement in observed prices.

As for the game-level analysis, a game listing on Epic has significant effects on price and discount depth within specific time windows. Both the stacked and synthetic

control difference-in-differences models find a significant increase in price within a 90 day window around the listing date, at ~\$0.50 on the Steam price of that same title. Within this window the estimate is inconsistent with downward competitive pressure on the Steam price. The second significant result comes from the synthetic control model alone, which finds a ~1.2 pp increase in discount depth within 180 days of the listing. Together, these two results suggest that list prices rose while deeper discounting partly offset the increase, since net price at 180 days is flat and insignificant even as discount depth rises. These are the only significant estimates across the three models, so for all other windows it cannot be confidently stated that an Epic listing affected the Steam price or discount depth of the same title. Across all three models, no detectable effect on Steam sale frequency appears when Epic lists the same game.

7.2 Hypothesis I: Short-Run Incumbent Insulation

The first and central hypothesis of this study states that a game listing on Epic has no effect on that same title's price or sale composition on Steam in the short run. Direction matters more than magnitude for evaluating this hypothesis. Epic's listing did not push Steam prices down. Within 90 days of a listing, both difference-in-differences models estimate that the Steam price of the same title rose by ~\$0.50, and this finding is statistically significant. Sale frequency returns null estimates in every window under all models, with small and inconsistent estimates. Discount depth returns a single significant estimate from the synthetic control model, with a 1.2 pp increase within 180 days.

Three important qualities of these findings are worth mentioning. First, only three coefficients reach statistical significance across the thirty estimates produced by the three models. Overwhelmingly, these models indicate that Epic's listings had a statistically

insignificant effect on the pricing and discounting behavior on Steam. Second, the statistically significant effects that do exist are small in magnitude. The 90-day window increase in price is ~\$0.50 and the 180-day window increase in discount depth is ~1.2 pp, which are both small in economic terms. Lastly, the direction of the 90-day price estimate indicates that Epic's listings did not force downward pressure on Steam prices. As a result, Steam listings show a small and inconsistent response to an Epic listing in the short run, and in no window does that response include falling prices.

7.3 Hypothesis II: Selection-Induced Differences

The second hypothesis of this study states that differences between Steam-only games and games on both platforms indicate compositional differences and selection into listing. The descriptive tables provide the strongest support for this claim. Comparing outcome variable values across the two groups shows clear and significant differences before Epic ever entered the market. In the pre-launch period, the groups differ significantly in price and sale frequency, though not in discount depth, with a \$4.45 nominal price difference that falls only to \$4.29 in real terms. Three release years (2014, 2015, and 2021) and three genres (adventure, open world, and narrative) also show significant differences between the groups. Although the market-level findings cannot substantiate the level differences due to the game fixed-effects absorbing any permanent gap between the groups, it does show that dual-platform prices fell substantially further than Steam-only prices between 2014 and 2018. This is consistent with two groups that differed in price from the start and converged over time. In addition to these findings, Jeong et al. (2026) find that Epic's selection of games for its promotional events, and by extension for its catalog, is not random. Collectively, the findings of this paper and Jeong

et al. (2026) indicate that games featured on both platforms differ from Steam-only games in terms of pricing and discounting behavior, and that this difference likely occurs from Epic's non-random selection of listings through promotional events and subsidization.

7.4 Hypothesis III: Long-Run Pricing Drift

The final hypothesis of this study states that pricing and discount behavior on Steam drift over the long run in a direction consistent with sustained competitive pressure from Epic. Although the market-level findings are not causal, the event-study model does produce significantly lower prices and higher sale frequency and discount depth for Steam games from 2022 through 2024. Therefore, the results are consistent with the claim that Epic's constant competitive pressure leads to changes in pricing and discounting behavior on Steam in the long run. I propose that the mechanism operates through Epic eroding the usage single-homing status of Steam users and capturing transactions through steering (Rochet & Tirole, 2003, 2006). One observable implication is that the response should be stronger for titles that list on Epic's platform. The price and discount depth interaction terms are significant from 2022 through 2024, which is consistent with that pattern, but the sale frequency interactions are only significant in 2024. The absence of a consistently significant game-level response is not in tension with this finding since the models in this section observe much smaller time windows and the significant market-level results occur years after Epic's entry. Any adjustment therefore appears to have operated gradually and through channels other than the individual listing decision. The findings of Jeong et al. (2026) on increased price elasticity and deeper discounts in longer time frames following Epic's promotional events coincide with this

claim that Epic's competitive pressure effects pricing and discounting behavior on Steam in the long run through eroding Steam's "local monopoly" on its singlehoming users (Armstrong, 2006).

7.5 Limitations and Future Research

Ultimately, this paper's largest limitation is the lack of quantity data. This forces the study to operate on the consumer side of the market when Epic's strongest competitive tool occurs on the seller-side of the market. Testing the impacts of Epic's entry on Steam revenue could uncover the true effect of the smaller revenue cut (12% vs 30%). In terms of the format of this consumer-side study, implementing a method that outputs stronger causal estimates of the market-level effect of Epic's entry would better validate the claim that pricing and discounting behavior on Steam drift in a direction consistent with Epic's competitive pressure eroding Steam's "local monopoly" on singlehoming users. Additionally, estimating a between-group specification with observable controls for genre, release date, and publisher, rather than game fixed effects, would allow the level difference between the two groups to be recovered directly. This would come at the cost of controlling for unobserved game characteristics, so the two approaches answer different questions rather than one improving on the other. Lastly, the 900-game sample used in this study contains the most popular games featured on Steam, which is not representative of Steam's full catalog and likely distorts the actual values of these outcomes. Applying this study to a more representative sample or differentiating between popular and less-popular games may produce different and more exploratory results.

As for future research, the obvious next step is implementing seller-side revenue data for Steam games, which would allow researchers to better understand the full effect of platform competition in the PC game market. Implementing better controls and improving the sample would strengthen the results. Lastly, classifying games based on the network externalities features like multiplayer access could provide better insight on why Steam's dominance remains after years of Epic's platform existing.

CHAPTER 8: CONCLUSION

The central question of this study was whether the entry of a well-funded platform that subsidizes both sides of the market affects consumer prices on a dominant platform in a two-sided market. This question was tested through the PC game market, where Epic entered with the intention of capturing market share from the incumbent platform, Steam, that dominated for 10+ years unrivaled. This entry was tested by collecting daily price and sale data for 900 Steam titles from 2013 to 2024 and examining Epic's impact at the market and game-level. At the game-level, games listing on Epic's platform led to little-to-no change in the pricing and discounting behavior of that same title on Steam in the short run. The only significant results were a ~\$0.50 increase in price within 90 days of the listing and a ~1.2 percentage point increase in discount depth within 180 days. As for the market-level findings, the results are consistent with Epic's constant competitive pressure creating downward pressure on prices and improving sale composition in the long run. These findings are consistent with two-sided market literature showing that users can usage-singlehome leading to the dominant platform having a local monopoly over its users which allows it to capture the bulk of transactions in the market. This study provides direct game-level evidence on whether entry affects consumer prices on a dominant platform, and provides clarity that an entrant's ability to obtain memberships does not automatically translate into capturing the transactions from the dominant platform.

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